Site Analysis: From Environmental Evidence to Design Response

ARCHITECTURE · architecture-architectural-design-and-site-planning · BoardPrep lesson v1

Official examination scope

The PRC June 2026 Architects program assigns 40% to Architectural Design and Site Planning for that cycle; it does not weight this lesson topic. This lesson combines original instruction with MIT course references. Confirm future program changes and applicable Philippine accessibility, safety, and building rules.

Learning objectives

- Build a site analysis from physical, environmental, access, use, and service evidence.
- Compare orientation, shading, daylight, and circulation against stated user and climate needs.
- Explain why a response follows from supplied evidence without making it a universal rule.

1. Map relationships before placing functions

Record contours, drainage, sun, prevailing wind, vegetation, views, noise, access, adjacent uses, hazards, and existing services. Separate measured conditions from assumptions. A boundary and north arrow are not a complete site analysis; program needs and surrounding context also shape design.

2. Translate evidence into comparisons

For each use, compare daylight, glare, heat, ventilation, privacy, noise, safety, and arrival. Orientation and shading are tools to test, not fixed answers. A shaded edge may support one use while another edge better serves access, views, or another condition.

3. Resolve trade-offs explicitly

State the user need, site constraint, proposed response, and remaining checks. Test effects on circulation, adjacent functions, outdoor space, and safe access. Apply current accessibility and building rules; a diagram does not demonstrate code compliance.

4. Communicate a reasoned proposal

Tie a program decision to observable site evidence and acknowledge trade-offs. A north-side reading room might reduce a stated west-glare concern, but still needs daylight, access, ventilation, and adjacency checks. Do not generalize one orientation to every building or climate.

Place a reading room from a site schedule

Question: The west edge has late-afternoon direct sun, the north edge is shaded, and pedestrians arrive from the east. A reading room needs usable daylight with limited direct glare.
1. Map the east arrival path and keep the entrance connection legible.
2. Compare a north-side room with west exposure and the daylight the room still needs.
3. Test shading, glazing, views, and circulation together.
Answer: Test a north-side location and shading against daylight, glare, and east-side arrival.
Check: The proposal follows the supplied schedule and is not a universal orientation rule.

Common misconceptions

- Treating compass direction as a complete design solution without climate or user evidence.
- Placing the entrance at geometric center while ignoring actual arrival paths.
- Choosing a flat location without checking safety, drainage, access, and use.
- Calling a diagram complete after mapping only the boundary and north arrow.

Practice patterns

- Choose which evidence to map before placing a building.
- Evaluate an entrance or function using actual arrival, climate, and use.
- Compare solar exposure and shade with daylight, glare, and circulation needs.

Official scope references

Official scope: Program for the June 8 and 10, 2026 Licensure Examination for Architects — Professional Regulation Commission; Professional Regulatory Board of Architecture — https://www.prc.gov.ph/sites/default/files/Program%20for%20June%202026%20LEA%28Architect%29_rev.pdf

Instructional references

Instructional reference: Introduction to Architecture and Environmental Design — MIT OpenCourseWare, Department of Architecture — https://ocw.mit.edu/courses/4-111-introduction-to-architecture-environmental-design-spring-2014/

Instructional reference: Environmental Technologies in Buildings — MIT OpenCourseWare, Department of Architecture — https://ocw.mit.edu/courses/4-401-environmental-technologies-in-buildings-fall-2018/

BoardPrep lesson material is original instruction. It is not an official examination document.
No claim about question frequency or examination weighting is made.